



Formal Languages and Finite Automata

Homework #3

1.

a.

$$S$$

$$S \rightarrow aSAB \mid aa$$

$$\Rightarrow aSAB$$

$$S \rightarrow aSAB \mid aa$$

$$\Rightarrow aaaAB$$

$$A \rightarrow B$$

$$\Rightarrow aaaBB$$

$$B \rightarrow bbB \mid bb$$

$$\Rightarrow aaabbB$$

$$B \rightarrow bbB \mid bb$$

$$\Rightarrow aaabbbbB$$

$$B \rightarrow bbB \mid bb$$

$$\Rightarrow aaabbbbb$$

b.

$$S$$

$$S \rightarrow aSAB \mid aa$$

$$\Rightarrow aSAB$$

$$B \rightarrow bbB \mid bb$$

$$\Rightarrow aSABbbB$$

$$B \rightarrow bbB \mid bb$$

$$\Rightarrow aSAbbbb$$

$$A \rightarrow B$$

$$\Rightarrow aSBbbbb$$

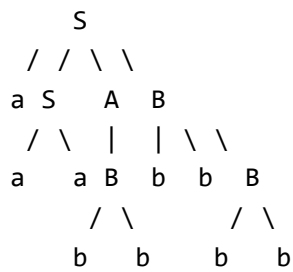
$$B \rightarrow bbB \mid bb$$

$$\Rightarrow aSbbbbbb$$

$$S \rightarrow aSAB \mid aa$$

$$\Rightarrow aaabbbbbbb$$

c.



2.

a.

$$S \rightarrow AB$$

$$A \rightarrow aA \mid \lambda$$

$$B \rightarrow bB \mid \lambda$$

b.

$$S \rightarrow aSc \mid A$$

$$A \rightarrow aAb \mid \lambda$$

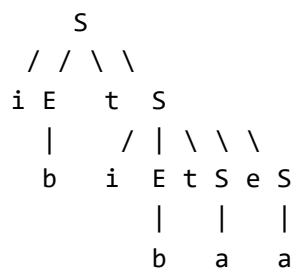
3.

Let

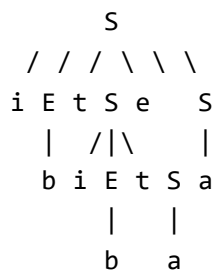
$$w = ibtibtaea$$

We can get two different trees.

a.



b.



They get the same result, but with different meanings.

4.

$$S \rightarrow aX$$

$$X \rightarrow aY$$

$$Y \rightarrow b \mid aXB$$

$$B \rightarrow b$$

5.

$$S \rightarrow AB \mid aA \mid bB \mid a \mid b \mid cC \mid c \mid d$$

$$A \rightarrow aA \mid bB \mid a \mid b$$

$$B \rightarrow bB \mid b$$

$$C \rightarrow cC \mid c \mid d$$

6.

a.

$$S \rightarrow X_a D_1 \mid X_a D_2$$

$$A \rightarrow X_b S \mid X_a X_a$$

$$B \rightarrow X_a B \mid b$$

$$C \rightarrow BA \mid X_b X_b$$

$$D_1 \rightarrow BA$$

$$D_2 \rightarrow X_b C$$

$$X_a \rightarrow a$$

$$X_b \rightarrow b$$

b.

$$S \rightarrow aBA \mid aX_b C$$

$$A \rightarrow bS \mid aX_a$$

$$B \rightarrow aB \mid b$$

$$C \rightarrow aBA \mid bA \mid bX_b$$

$$X_a \rightarrow a$$

$$X_b \rightarrow b$$